Small β range, i.e., β is $o\left(\frac{\log n}{\log d}\right)$.

Low rank approximation: $R = f_{k^*}(b \cdot A)$.

Since each entry of A is in $[-1, 1]$, each entry of $\beta \cdot A$ is in $[-\beta, \beta]$. But note that β in this case is $o\left(\frac{\log n}{d}\right) = O(\log n \cdot \Delta)$. By the definition of k^*, each entry of $\exp(\beta \cdot A) - f_{k^*}(\beta \cdot A)$ has absolute value $\leq \epsilon$. Therefore the overall error is $\leq \epsilon n$.

For sparse only: By assumption, $m = \Omega(\|L\|_0)$ entries in A are ≥ 0, which are exactly the entries in $\exp(\beta \cdot A)$ that are ≥ 1. Hence any (say) $\frac{m}{2}$ sparse approximation has error $\geq \sqrt{\frac{m}{2}} \geq \Omega(\sqrt{\|L\|_0})$. By our assumption, $\|L\|_0 = \Omega(n^2)$.

Mid-range β, i.e., $\beta \geq \frac{1}{l} \cdot \log n$ and β is $O(\log n)$.

Sparse only: the argument is the same as in the low β range.

Sparse + low-rank: The low-rank part $R = fst(\beta \cdot A)$. By Lemma 9, this has rank $n^{o(1)}$, so it has $n^{(1+o(1))}$ parameters.

The sparse part is $S = e^{\beta \cdot H} - R_{\mathrm{supp}(H)}$. Clearly this needs $|\mathrm{supp}(H)|$ parameters.

Let $E = M_\beta - (S + R)$. Then (i) in $\mathrm{supp}(H)$, E is all 0. (ii) output of $\mathrm{supp}(H)$, by definition, entries of $\beta \cdot A$ are in $[-\beta\Delta, \beta\Delta]$, which in the current range of β is $[-O(\log n\Delta), O(\log n\Delta)]$. Therefore all the entries of E have absolute value $\leq \epsilon$. By the definition of k^*, we have that $\|E\|_F \leq \epsilon n$.

Low-rank only: Let $\widetilde{R}$ be rank $r - n^{o(1)} - 1$ that approximates M_β. Then using the same argument as our existing lower bound argument, we get that $\widetilde{R} - R \approx_E S$ (this means that the error $\leq \|E\|_F + \left\|M_\beta - \widetilde{R}\right\|_F$). Now note that $S = e^{\beta \cdot H} - (f_{k^*}(\beta \cdot A))_{\mathrm{supp} H}$ is a symmetric, block diagonal matrix with $r = \Omega(n)$ blocks. Corollary 7 implies that $\lambda_r(S)$ is at least the smallest non-diagonal value in S. Now the smallest non-diagonal value in $e^{\beta \cdot H}$ is $\geq e^{\frac{1}{l}\log n} = n$. On the other hand, the largest value in $(f_{k^*}(\beta \cdot A))_{\mathrm{supp} H}$ is

$$
\begin{aligned}
&\leq k^* \frac{\beta^{k^*}}{k^*!} \leq \beta \cdot \left(\frac{e\beta}{k^* - 1}\right)^{k^*-1} \\
&\lesssim \log n \left(\frac{e \cdot \log n}{\log n \cdot \Delta}\right)^{O(\log n \cdot \Delta)} \\
&\lesssim \log n e^{O(\log n \cdot \Delta \cdot \log \frac{1}{\Delta})} \\
&\lesssim \log n \cdot n^{o(1)} \\
&= n^{o(1)}.
\end{aligned}
$$

Hence $\lambda_r(S)$ is $\Omega(n)$. The claimed result then follows since $\|E\|_F \leq \epsilon n$ and $\mathrm{rank}\widetilde{R} - R \leq r - 1$ (Eckart-Young-Mirsky theorem).

Large β range, i.e., $\beta \geq \omega(\log n)$.

Sparse only: $S = e^{\beta \cdot H}$. Note that each entry in $E = M_\beta - S$ is upper bounded by $e^{\Delta \cdot \beta} \leq e^{o\left(\frac{\beta}{\log d}\right)}$. Then

$$
\begin{aligned}
\|E\|_F &\leq n \cdot e^{o\left(\frac{\beta}{\log d}\right)} \\
&\leq \epsilon \cdot e^{\log \frac{n}{\epsilon} + o\left(\frac{\beta}{\log d}\right)} \\
&\leq \epsilon \cdot e^{o(\beta) + o\left(\frac{\beta}{\log d}\right)} \\
&\leq \epsilon \cdot e^{o(\beta)} \\
&\leq \epsilon \cdot e^{\beta / l}.
\end{aligned}
$$

Low-rank only: since $\|E\|_F$ is $\leq \epsilon e^{\beta/l}$, it is enough to argue that any rank r-approximation to S has error $\geq e^{\beta/l}$. But the latter follows since $\lambda_r(S) \geq e^{\beta/l}$. This is because $e^{b \cdot H}$ is symmetric and each entry in H is $\geq \frac{1}{\lambda}$. Then we can use Corollary 7. Eckart-Young-Mirsky then completes the proof. $\square$

D.3 Scatterbrain: Analysis

Here we prove Theorem 2, which shows that Scatterbrain approximation is unbiased and analyses its variance. We restate the theorem here for the reader's convenience.

Theorem. *Define $\sigma(q,k) = \exp(q^\top k)$, $\widehat{\sigma}^{\mathsf{pfe}}$ as Performer's estimator and $\widehat{\sigma}^{\mathsf{sbe}}$ as Scatterbrain estimator. Denote $\mathcal{S}^{d-1} \subset \mathbb{R}^d$ as the unit sphere. Suppose $q, k \in S^{d-1}$ are such that $\|q-k\| < \tau$. Then:*

$$\mathbb{E}[\widehat{\sigma}^{\mathsf{sbe}}(q,k)] = \sigma(q,k), \quad \mathrm{Var}[\widehat{\sigma}^{\mathsf{sbe}}(q,k)] = (1-p)\cdot \mathrm{Var}[\widehat{\sigma}^{\mathsf{pfe}}(q,k)] < \mathrm{Var}[\widehat{\sigma}^{\mathsf{pfe}}(q,k)]$$

where $p = \exp(-\frac{\tau^2}{4-\tau^2}\ln d - O_\tau(\ln\ln d))$.

Proof. Let $A_{ij} = \exp(q_k^\top k_j)$ be ij-entry of the unnormalized attention matrix, $A^{\mathrm{lr}}_{ij} = \phi(q_i)^\top \phi(k_j)$ the entry of the low-rank approximation (Performer), and let A^{sb}_{ij} be the entry of the Scatterbrain (sparse + low-rank) approximation. By the construction of the Scatterbrain attention matrix (Eq. (1)), if $ij \in \mathcal{S}$, where $\mathcal{S}$ is the set of indices selected by the LSH, then:

$$A^{\mathrm{sb}}_{ij} = (\widetilde{Q}\widetilde{K}^\top + S)_{ij} = \phi(q_i)^\top \phi(k_j) + \exp(q_i^\top k_j) - \phi(q_i)^\top\phi(k_j) = \exp(q_i^\top k_j).$$

If $ij \notin \mathcal{S}$, then

$$A^{\mathrm{sb}}_{ij} = (\widetilde{Q}\widetilde{K}^\top + S)_{ij} = \phi(q_i)^\top \phi(k_j) + 0 = \phi(q_i)^\top \phi(k_j).$$

In other words, A^{sb} matches A on the indices in $\mathcal{S}$, and matches A^{lr} on the indices not in $\mathcal{S}$.

To show that A^{sb} is an unbiased estimator of A, we simply use the fact that A^{lr} is also an unbiased estimator of A [17, Lemma 1]:

$$\begin{aligned}
\mathbb{E}[A^{\mathrm{sb}}_{ij}] &= \mathbb{P}(ij \in \mathcal{S})\mathbb{E}[A_{ij} \mid ij \in \mathcal{S}] + \mathbb{P}(ij \notin \mathcal{S})\mathbb{E}[A^{\mathrm{lr}}_{ij} \mid ij \notin \mathcal{S}] \\
&= \mathbb{P}(ij \in \mathcal{S})A_{ij} + \mathbb{P}(ij \notin \mathcal{S})A_{ij} \\
&= A_{ij}.
\end{aligned}$$

In other words, $\mathbb{E}[\widehat{\sigma}^{\mathsf{sbe}}(q,k)] = \sigma(q,k)$.

Now we analyze the per-entry variance of A^{sb}. Since A^{sb} is an unbiased estimator of A, by the law of total variance,

$$\begin{aligned}
\mathrm{Var}(A^{\mathrm{sb}}_{ij}) &= \mathbb{P}(ij \in \mathcal{S})\mathrm{Var}(A_{ij} \mid ij \in \mathcal{S}) + \mathbb{P}(ij \notin \mathcal{S})\mathrm{Var}(A^{\mathrm{lr}}_{ij} \mid ij \notin \mathcal{S}) \\
&= \mathbb{P}(ij \in \mathcal{S}) \cdot 0 + \mathbb{P}(ij \notin \mathcal{S})\mathrm{Var}(A^{\mathrm{lr}}_{ij}) \\
&= \mathbb{P}(ij \notin \mathcal{S})\mathrm{Var}(A^{\mathrm{lr}}_{ij}).
\end{aligned}$$

To compute the probability that the index ij is not in $\mathcal{S}$ (i.e., not selected by LSH), we use the standard bound on cross-polytope LSH [3, Theorem 1]:

$$p := \mathbb{P}(ij \in \mathcal{S}) = \exp(-\frac{\tau^2}{4-\tau^2}\ln d - O_\tau(\ln\ln d)).$$

Therefore,

$$\mathrm{Var}(A^{\mathrm{sb}}_{ij}) = (1-p)\mathrm{Var}(A^{\mathrm{lr}}_{ij}) < \mathrm{Var}(A^{\mathrm{lr}}_{ij}).$$

In other words, $\mathrm{Var}[\widehat{\sigma}^{\mathsf{sbe}}(q,k)] = (1-p)\cdot\mathrm{Var}[\widehat{\sigma}^{\mathsf{pfe}}(q,k)] < \mathrm{Var}[\widehat{\sigma}^{\mathsf{pfe}}(q,k)]$.

More explicitly, by plugging in the variance of A^{lr} [17, Lemma 2], we have

$$\mathrm{Var}(A^{\mathrm{sb}}_{ij}) = (1-p)\frac{1}{m}\exp\left(\|q_i + k_j\|^2\right)\exp(2q_i^\top k_j)\left(1 - \exp\left(-\|q_i+k_j\|^2\right)\right),$$

where $p = \exp(-\frac{\tau^2}{4-\tau^2}\ln d - O_\tau(\ln\ln d))$

$\square$

E Additional Experiments and Details

E.1 Datasets

ImageNet [25]: ImageNet is one of the most widely-used image classification benchmarks. In our experiments in Section 5.1 of evaluating the approximation accuracy of Scatterbrain, both BigGAN and Vision Transformer are pre-trained on this dataset. It has roughly 1.2 million training images and 50,000 validation images.

WikiText103 [45] and Copy [36]: WikiText103 is a popular dataset for auto-regressive models. It is from a collection of over 100 million tokens extracted from the set of verified good and featured articles on Wikipedia. It has 28,475 training articles, 60 for validation and 60 for testing.

Copy is a synthetic a synthetic sequence duplication task where inputs are of the form $0w0w$ and $w \in \{0, ..., N\}^*$. It is previously used in [15, 36]. This task is useful for demonstrating the effectiveness of long range attention: it requires non-local attention lookups. It cannot be solved by any model relying on sparse attention with a limited range such as, local attention.

Long Range Arena (LRA) [57]: This is a recent benchmark for evaluating efficient transformers with long input sequence. We used ListOps [46], byte-level IMDb reviews text classification [44], byte-level document retrieval [48], image classification on sequences of pixels [37] and Pathfinder [40]. We follow the same evaluation mechanism from [57] but implement our own version in Pytorch (like data loader).

GlUE [64]: GLUE is a standard multi-task benchmark in NLP. It has single-sentence tasks, CoLA and SST-2; similarity and paraphrase tasks, MRPC, STS-B, QQP; and inference tasks, MNLI, QNLI, RTE and WNLI. For our additional experiments below (not enough space to be included in the main paper), we follow the tradition from [22, 26, 68] and truncate all the input sequences to 128 tokens.

E.2 Settings

BigGAN: We adapt the same pre-trained BigGAN model from [22] with no additional training. The model has a single attention layer at resolution 64×64 (4096). Similar to the prior work, we also replace its full attention layer with Scatterbrain at the same resolution. Figure 5 in the main paper shows the best-effort comparison with [1/32, 1/16, 1/8, 1/4, 1/2] of the parameter budget. For example, if given parameter budget 1/2, we report the best performance of Smyrf from choice of 32/64/128 hash round 64/32/16 cluster size.

T2-ViT: We use the pre-trained vision transformer model T2T-ViT-14 from [69] with 224×224 image size. Without any additional training, we just replace the attention layer with Scatterbrain and other baselines and evaluate the approximation error and classification accuracy on ImageNet testings. Again, we report the best-effort best performance of each approximation given the certain parameter budget.

Auto-regressive Model: We follow the settings from the popular repo `https://github.com/NVIDIA/DeepLearningExamples` for training vanilla Transformer from scratch on WikiText103, except for chunking WikiText103 into sequence length 1024 in order to simulate long input sequences. The model is 16 layer with 8 head and 512 model dimension. We train all the models for 30 epochs and report the best Testing Perplexity. The model we use for Copy task is simply a 2-layer-4-head transformer and sequence length is also 1024. We make 5 runs and report average. Table 4 presents the results with standard deviation.

Classification Model: We follow the model setting from [57, 67]. We share the same finding with [67] that the acuracy for the Retrieval tasks is actually higher than reported in [57].

Ratio between Sparse and Low-rank components: There are some rules that we used in our experiments to set this ratio. For inference, we set this ratio based on the entropy of an observed subset of attention matrices in different layers: we allocate more memory to the low-rank component compared to the sparse component if the entropy is high. For training, generally allocating more memory budget to sparse tends to perform better, so in the experiment, we set the ratio to 3:1 (sparse: low-rank component) for simplicity. Moreover, in future work, it could be useful to make this ratio adaptive during training. For example, in the early stage of the training and early layers, attention matrices are usually more uniform (higher entropy). Thus, the approximation error could be even lower if the ratio favors low-rank-based components. One approach could be to monitor the approximation error of sparse and low-rank components compared to full attention regularly and adjust the memory budget accordingly. We will add the above discussion to the updated manuscript.

Table 4: The performance of Scatterbrain, REFORMER, PERFORMER and Full-Attention on Long-Range-Arena benchmarks and 2 popular language modeling tasks. We fix the same number of parameters (1/8 of the full) used for approximating the attention matrix for each method.

Attention	Copy (ppl)	WikiText-103 (ppl)
Full Attention	1	25.258±0.37
Reformer	6.8±0.64	27.68±0.53
Performer	49±2.7	66±5.8
Scatterbrain	**2.58**±0.21	**26.72**±0.44

Attention	ListOps	Text	Retrieval	Image	Pathfinder	Avg
Full Attention	38.2±0.17	63.29±0.38	80.85±0.12	41.78±0.26	73.98±0.31	59.62
Reformer	36.85±0.37	58.12±0.42	78.36±0.29	28.3±0.39	67.95±0.28	53.9
Performer	35.75±0.29	62.36±0.49	78.83±0.33	39.71±0.48	68.6±0.36	57.05
Scatterbrain	**38.6**±0.22	**64.55**±0.34	**80.22**±0.31	**43.65**±0.46	**69.91**±0.25	**59.38**

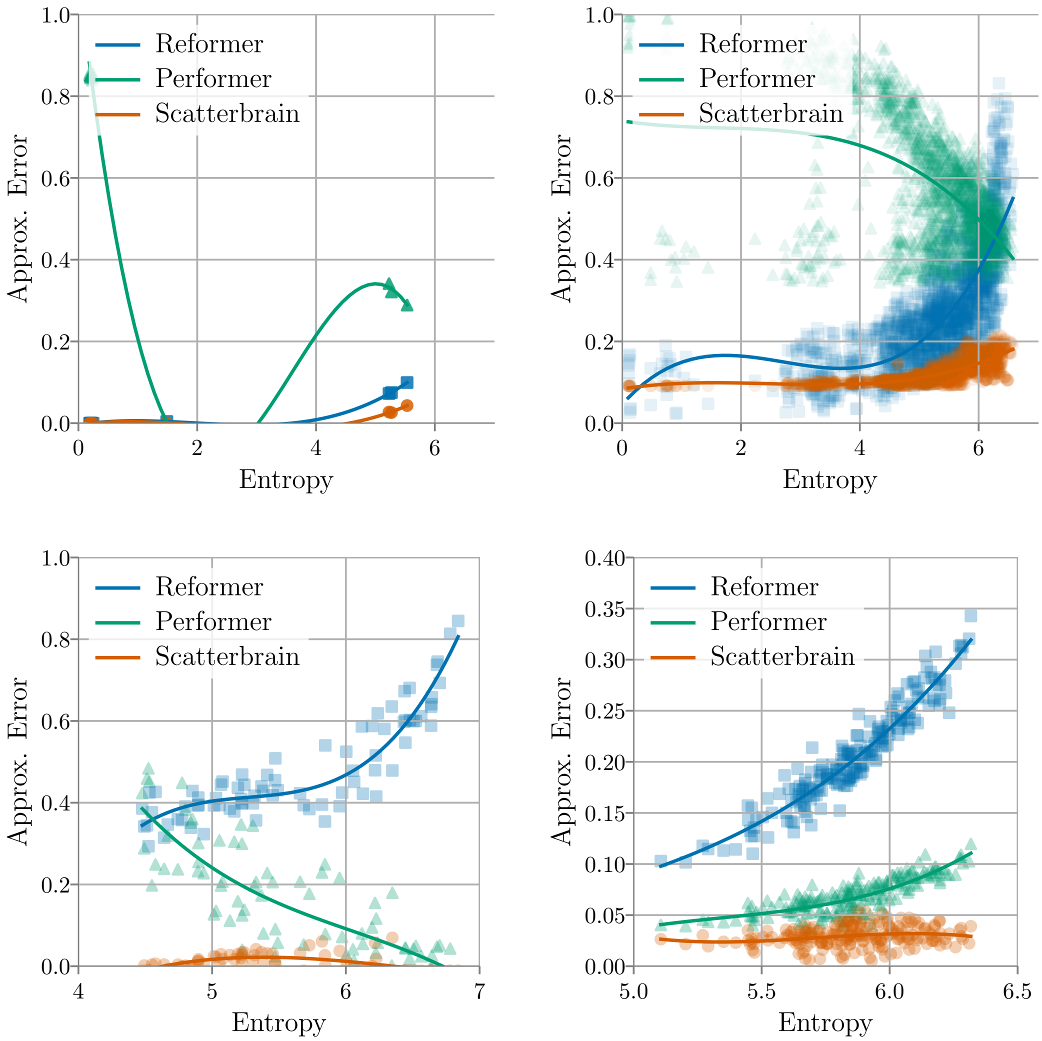

Figure 8: Top two plots present Approximation Error vs. Entropy of attention matrices for REFORMER, PERFORMER and Scatterbrain on Copy (left) and WikiText103 (right). Bottom two plots present Approximation Error vs. Entropy of attention matrices for REFORMER, PERFORMER and Scatterbrain on Text-IMDb (left) and Image-Cifar10 (right). Recall we observe that entropy of the softmax attention distribution (i.e., scale of logits) determines the regimes where sparse, low-rank, or sparse + low-rank perform well. Scatterbrain yields better approximation than REFORMER or PERFORMER in most of the cases; PERFORMER performs the worst on language modeling tasks while REFORMER performs the worst on classification tasks. These plots for approximation error analysis match with their performance on downstream tasks.